

Computer-Vision-Based ROBOTIC ARM

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The system is designed to work like a roadside shopkeeper. When a customer places an order such as 1kg of apples, cameras connected to a GPU-enabled computer identify the required items and determine their location using computer vision.

The robotic arm, positioned at the centre of the shop for maximum reach, receives the target coordinates via Wi-Fi. It moves to the identified location, picks up the required quantity, and measures the weight using a load cell and distance sensor mounted on the gripper. It then places the items on a weighing tray. For an order such as 1kg of apples, the arm may make multiple pickup trips until the required weight is reached, after which the items can be packaged and delivered.

Object detection can be handled efficiently with modern AI models such as YOLO and specialised models like YOLOE for custom or uncommon objects. Since AI-based vision and path planning require significant computing power, they are handled by a GPU-enabled computer, laptop, or Raspberry Pi 5, while an ESP32 manages real-time hardware control.

Hybrid architecture

The system uses a hybrid architecture in which a GPU-enabled computer or Raspberry Pi 5 acts as the high-level processing unit. It performs computationally intensive tasks such as object detection, AI-based decision-making, and path planning. The resulting commands are transmitted to the ESP32 through Wi-Fi, Ethernet, or serial communication, depending on the application and system configuration.

The ESP32 acts as the real-time control unit, interfacing directly with the motors, sensors, encoders, and gripper. It executes the commands received from the high-level computer and provides precise, responsive control of the robotic mechanism. The architecture therefore combines high-level AI processing with real-time hardware control. Fig. 1 shows the authors' working prototype. The components required to build the system are listed in the Bill of Materials table.

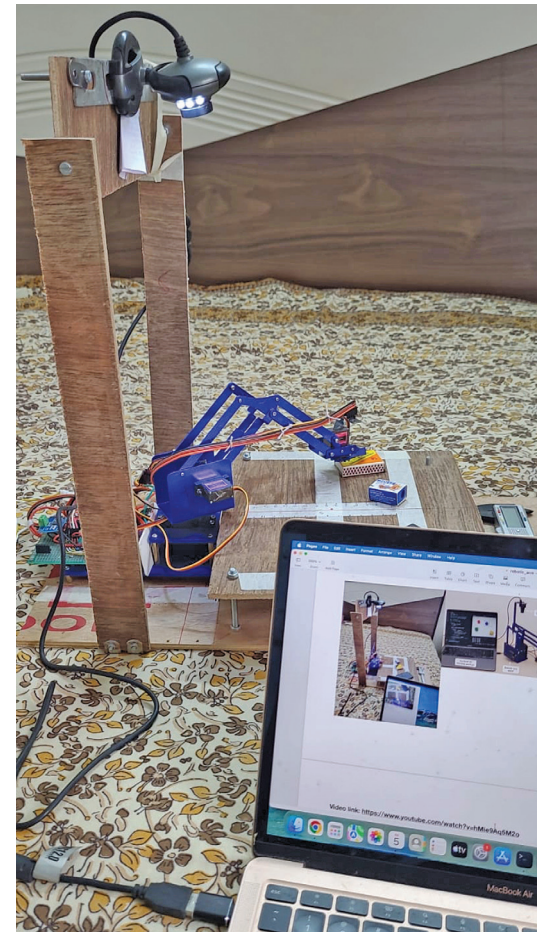


Fig. 1: Authors' prototype

BILL OF MATERIALS

Component	Description	Quantity
ESP32 development board	Wi-Fi-enabled microcontroller for hardware control	1
PCA9685 PWM driver	16-channel servo driver board	1
Hobby servo motors	Standard 180° pan-and-tilt servos (expandable)	5
VL53L0X ToF sensor	Time-of-flight distance sensor	1
128×64 OLED display	SH1106 or SSD1306 I²C OLED module	1
USB webcam	External camera mounted on the pan-and-tilt assembly	1
INA219 current sensor	Current and voltage monitoring sensor	1
100μF capacitor	Smoothing capacitor across the 5V rail	1
ATX power supply	Power supply unit with connector	1
Connecting wires	Assorted jumper wires and cables	As needed
Robot arm and claw kit	Mechanical arm and claw assembly	1

Mechanical design and arm operations

The mechanical design of the robotic arm is critical to the system's performance. It must provide adequate reach, payload capacity, positioning accuracy, stability, and workspace coverage while remaining cost- and